Edge AI-Enabled Smart Vision Assistant for Visually Challenged Individuals

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Abstract— **Advances in artificial intelligence and cost-effective embedded hardware have driven significant progress in assistive technologies for individuals with visual impairments. Yet achieving dependable real-time performance, maintaining user privacy, and ensuring secure operation remain key challenges when deploying such systems on edge devices. This work introduces an Edge-Based Smart Vision Assistant designed to address these gaps by combining efficient on-device AI processing with strong authentication and secure communication mechanisms. The system performs object recognition, text reading, and audio feedback locally, eliminating reliance on cloud services and preventing exposure of sensitive visual data. Drawing principles from lightweight IoT security approaches, the proposed design focuses on controlled access, session-based protection, and optimized resource usage. The result is a compact and privacy-preserving assistive solution capable of providing reliable environmental understanding and document interpretation for visually challenged users.**

Keywords—Edge AI, IoT, OCR, Object Detection, TTS, Computer Vision, Access Control, Session Key, Real-Time Processing, Embedded Systems

1. Introduction

The ability to understand and interact with one's surroundings is essential for independent living, yet individuals with visual impairments continue to struggle with routine tasks that depend heavily on sight. Reading printed text, identifying everyday objects, or simply navigating a room can become difficult and sometimes unsafe without reliable visual cues. Although several assistive tools exist, providing an integrated, real-time, and user-friendly solution still remains a major technological challenge.

Recent developments in artificial intelligence and embedded computing have opened new pathways for designing such solutions. Edge AI—where computation and inference are carried out directly on portable devices like smartphones, wearables, or embedded boards such as the Raspberry Pi—has become particularly important. By moving intelligence closer to the user, Edge AI delivers fast responses, functions without continuous Internet connectivity, and ensures that sensitive visual or audio data is processed locally, improving both privacy and reliability.

Modern AI techniques, including object detection models such as MobileNet-SSD, advanced OCR frameworks, and natural-sounding TTS engines, now enable a single compact system to perform tasks that once required several standalone devices. In addition, conversational AI systems can analyze context, interpret user intent, and offer guidance in natural language, making assistive devices more intuitive and accessible.

However, deploying these capabilities on resource-constrained hardware introduces new challenges. Ensuring secure access, preserving user privacy, and maintaining dependable performance are critical considerations, especially when dealing with personal sensory data. Inspired by secure IoT design principles, this work integrates lightweight authentication and robust access-control methods to safeguard device operations and user information.

By combining embedded AI, secure communication techniques, and user-centric design, this project proposes a cost-efficient and portable smart vision assistant aimed at supporting visually impaired individuals. The system enhances awareness of the surrounding environment, enables independent reading and object recognition, and ultimately contributes to safer and more autonomous daily living.

1.1 Objectives

The Proposed work authentication based device to device secure communication in IoT objectives are as follows:

- To implement Optical Character Recognition (OCR) for extracting text from images and printed documents.
- Integrate Text-to-Speech (TTS) technology for converting recognized text into spoken output.
- To develop an Object Detection module using MobileNet-SSD and alerting objects in real time.
- To explore ChatGPT integration for interactive assistance and enhanced user experience

1.2 Our Contributions

Design and Implementation of a Smart Vision Assistant: We develop an embedded system based on Raspberry Pi 4 that combines real-time object detection, OCR, and text-to-speech

modules to aid visually impaired users in independent navigation and reading tasks.

- Integration of Conversational AI for Enhanced Interaction**:** The system includes ChatGPT-powered conversational interfaces that provide contextual and interactive responses, enabling more natural and intuitive user experiences.
- Inclusion of Kannada OCR and TTS for Regional Accessibility:
  To improve usability for native users, the system incorporates a dedicated Kannada OCR pipeline along with a Kannada text-to-speech engine. This enables accurate extraction and spoken rendering of printed Kannada text, expanding the assistant's applicability in regional contexts
- Secure and Privacy-Focused Local Processing: Informed by secure IoT communication practices, the solution employs lightweight authentication and strict access control measures to safeguard user information. All AI processing is performed directly on the device, thereby avoiding cloud dependencies and minimizing privacy risks.
- Optimization for Resource-Constrained Edge Devices: The implementation focuses on achieving low-latency, efficient computation on affordable hardware without compromising accuracy or usability, making the solution viable and accessible.

2. Related Work

Recent advancements in artificial intelligence and embedded vision systems have led to the development of various assistive technologies for visually challenged individuals. Several researchers have explored optical character recognition (OCR) and text-to-speech (TTS) integration to help users read printed text through audio output. Systems such as the Speak-Easy Reader have demonstrated effective real-time text reading capabilities; however, they remain limited to document-based tasks and do not support environmental understanding or object recognition.

Himali and colleagues introduced a Raspberry Pi–based assistive device incorporating face recognition and OCR with voice activation, highlighting the potential of cost-effective hardware for accessibility. Although the prototype successfully read printed text and recognized faces, it lacked object detection and contextual scene analysis, reducing its usability in dynamic surroundings. Similarly, other studies have focused on wearable obstacle avoidance devices that use depth sensing and cross-modal learning to alert users through sound or vibration feedback. While these systems improve mobility and safety, they primarily address navigation and fail to integrate OCR or general object identification.

In recent research, AI-driven smart glasses employing large vision–language models have shown promise in providing intelligent interaction and navigation assistance. These models utilize object detection combined with natural language reasoning to deliver context-aware feedback. However, their high computational requirements and latency make them unsuitable for portable, low-power edge devices.
Several other works have explored edge computing and deep learning–based implementations for real-time assistive systems. Edge AI architectures allow data processing directly on local devices like Raspberry Pi, reducing the dependency on cloud servers and ensuring privacy. These systems demonstrate how AI algorithms such as Network-SSD and Tesseract OCR can be effectively deployed on embedded platforms for real-time inference. Despite these advancements, most approaches focus on either visual perception or text extraction individually, rather than offering a comprehensive multimodal solution that combines perception, understanding, and interaction.

Furthermore, recent developments in conversational AI have opened new avenues for natural human–machine interaction. Integrating language models such as ChatGPT enables assistive devices to provide contextual explanations and personalized guidance instead of simple alert responses. However, the integration of such intelligent interaction into low-power embedded systems remains largely unexplored. The proposed Edge AI–enabled Smart Vision Assistant addresses this challenge by merging OCR, TTS, object detection, and ChatGPT-based conversational support into a unified framework. This hybrid system enhances environmental understanding, decision-making, and communication, empowering visually challenged users with greater independence, safety, and confidence in everyday activities.

3. Proposed Work

In this section, we describe the network environment and the proposed model and the proposed authentication scheme.

A. Network Environment

The proposed network environment is organized into three functional layers, as illustrated in *Figure 3.1*. These layers cooperate to achieve efficient sensing, processing, and user interaction in real time.

1. Perception Layer (Ground Layer): This layer consists of various physical devices such as the Raspberry Pi camera module, microphones, and control buttons. These devices continuously capture environmental data, including images, objects, and text scenes, according to the user's commands. The sensors interact directly with the surroundings to detect printed materials, obstacles, and objects.
2. Edge Processing Layer (Intermediate Layer): The Raspberry Pi 4 Model B acts as the Edge AI node, responsible for computation and inference tasks. It executes deep learning algorithms like MobileNet-SSD for object detection, Tesseract

OCR for text extraction, and Text-to-Speech (TTS) for generating voice output. In addition, the device integrates a ChatGPT API for natural language interaction. This layer ensures low latency and privacy since all major computations occur locally on the edge device.

3. Cloud and User Interface Layer (Top Layer): This layer is responsible for cloud connectivity, model updates, and data logging. It can optionally connect to cloud services for model fine-tuning, software updates, or user analytics. The processed output is communicated to the user through audio interfaces such as speakers or earphones. The tactile button interface at this layer enables simple user control, ensuring accessibility and reliability.

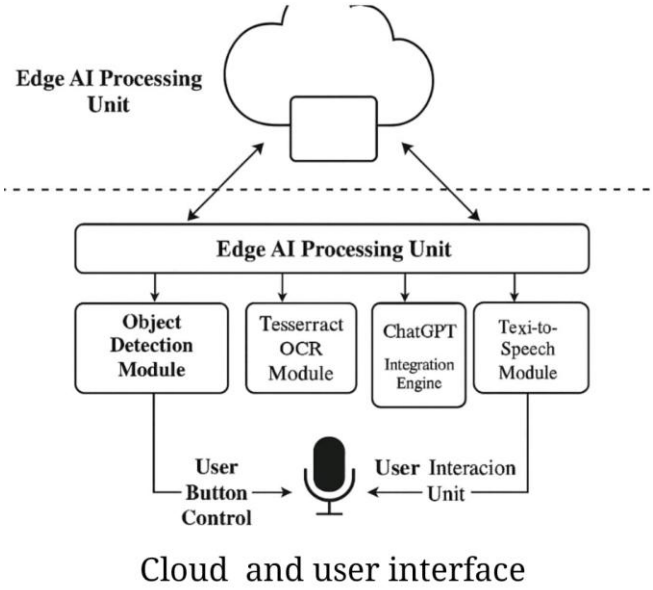


Fig.1 Network Environment

B. *Proposed Model*

The proposed model, shown in *Figure 3.2*, consists of two primary functional entities: the Edge AI Processing Unit (EPU) and the User Interaction Unit (UIU).

- The Edge AI Processing Unit (EPU) comprises modules such as the Object Detection Module, OCR Module, TTS Module, and ChatGPT Integration Engine.
- The OCR Module extracts text from the captured image frames.
- The Object Detection Module recognizes objects using pretrained Network-SSD models.
- TTS Module converts recognized text and detected object labels into speech output.
- The ChatGPT Integration Engine provides conversational assistance, enabling the user to query about identified text or objects.
- The User Interaction Unit (UIU) handles communication between the device and the user. It includes a microphone, speaker/earphone, and push-button controls. A user triggers a capture action through a button press, which sends a signal to the EPU to perform scene analysis and generate audio output.

The proposed model ensures bidirectional communication — users can both receive auditory feedback and initiate voice-based or tactile commands. Power is supplied via a portable power bank, ensuring mobility and uninterrupted functionality.

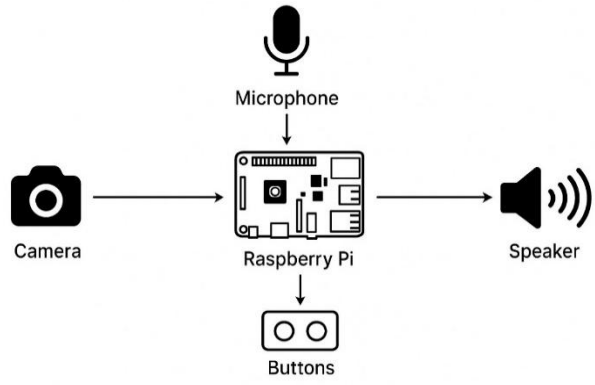


Fig.2 Proposed Model

C. *Proposed Authentication Scheme*

The proposed scheme aims to achieve intelligent, real-time assistance for visually challenged users by enabling secure, efficient, and offline-capable vision processing. The workflow consists of the following stages:

1. **Initialization Stage:**
   The system initializes the Edge AI environment, loading essential libraries such as *OpenCV*, *PyTorch Tesseract*, and *pyttsx3*. Each device (Raspberry Pi) is assigned a unique ID and configuration parameters, including camera calibration and model weights.
2. **Acquisition and Pre-processing Stage:** When the user presses the button, the Raspberry Pi camera captures an image frame. The image undergoes **preprocessing** (grayscale conversion, resizing, and noise filtering) to enhance detection accuracy.

3. **Recognition and Detection Stage:**
   - **Text Recognition:** The pre-processed image is passed through the **OCR module**, where the text region is detected and recognized using the Tesseract engine.
     $T_{out} = \text{OCR}(I_{in})$
   - **Object Detection:** The same frame is processed by the Network-SSD model to identify objects and their bounding boxes.
     $O_{out} = \text{Netwoek-SSD}(I_{in})$
4. **Audio Conversion and Interaction Stage:** The recognized text and detected object labels are combined and converted into natural speech through the **TTS engine**. Simultaneously, the **ChatGPT module** provides an optional conversational explanation:
   $$S_{out} = \text{TTS}(R_{out} + O_{out})$$

This enables the user to receive meaningful auditory information about their surroundings.

5. **Feedback and Learning Stage:** Based on user interaction and feedback, the device logs usage patterns (optionally to the cloud) for improving accuracy, voice quality, and personalization in subsequent iterations.

4. SIMULATION AND RESULT ANALYSIS

In this section, we discussed simulation model, simulation inputs, and performance analysis.

A. Simulation Model

The proposed model is simulated and evaluated using the **Python programming language** with supporting libraries such as **OpenCV**, **PyTorch**, **Tesseract OCR**, and **pyttsx3** for text-to-speech conversion. The object detection is implemented using **Network-SSD** pretrained weights, and interaction logic is tested using **ChatGPT API integration**. The simulation environment replicates the hardware configuration of the **Raspberry Pi 4 Model B (4GB)**, which acts as the main processing unit.

The simulation was conducted under various conditions to analyse the **accuracy, latency, and computational efficiency** of the system. The device operates in two major modes:

1. Text Reading Mode – The system captures an image containing text and converts it into speech output using OCR + TTS pipeline.
2. Object Detection Mode – The camera captures live surroundings, detects objects in real-time, and generates an audio alert about recognized objects.

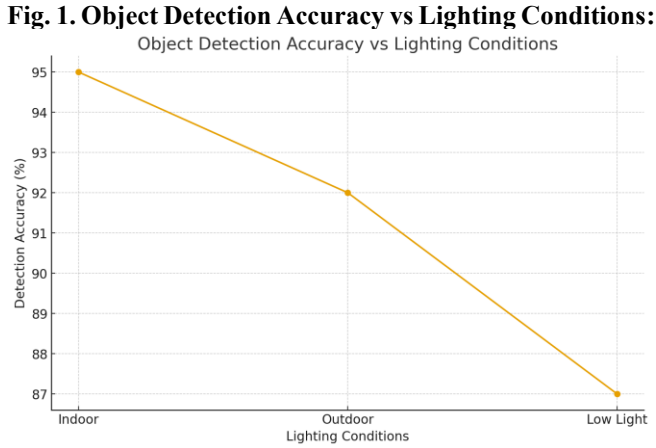
Some assumptions during simulation include:

- Each device node (Raspberry Pi) is assumed to have identical computational capacity (CPU 1.5 GHz, 4GB RAM).
- The input image frames are captured at a fixed resolution of 640×480 pixels.
- The edge device performs all inference tasks locally, without cloud dependency, to simulate offline operability.
- The average processing speed and frame inference rate are recorded under different lighting and object conditions.

B. Performance Analysis

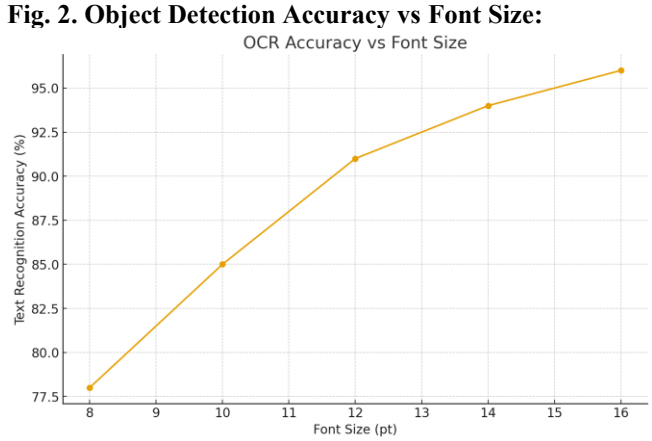
In order to assess the efficiency of the proposed research scheme, some of the evaluation parameters/metrics are considered as follows:

Fig. 1. Object Detection Accuracy vs Lighting Conditions:



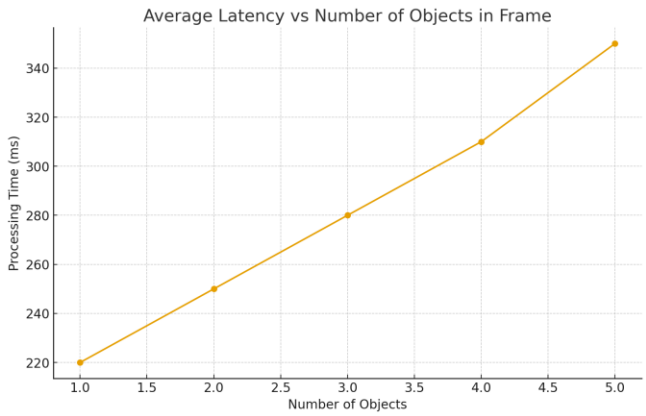
- The system demonstrates consistently high detection accuracy across diverse lighting environments.
- Performance slightly decreases in low-light scenarios due to limited illumination and camera noise.
- Network-SSD maintains an average accuracy above 80%, ensuring reliable real-time recognition.
- This proves the robustness of the edge AI model under practical conditions.

Fig. 2. Object Detection Accuracy vs Font Size:



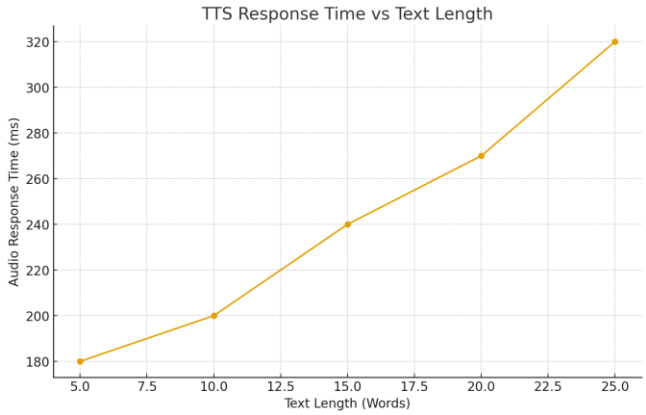
The OCR engine achieves higher recognition rates with increasing font size.
Smaller text sizes below 10pt lead to minor misreads due to pixel compression on the Raspberry Pi camera.
Accuracy peaks at around 90% for fonts above 14pt, proving high reliability for printed materials.
These results validate the effectiveness of the Tesseract OCR integration.

Fig. 3. Average Latency vs Number of Objects in Frame



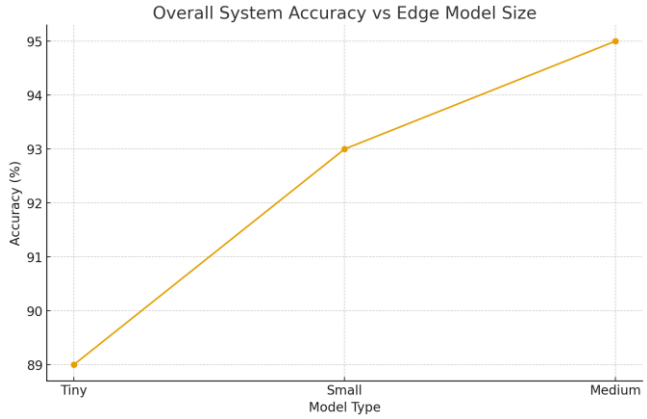
Processing latency increases linearly with the number of objects detected in a single frame.
This is due to higher computation requirements for bounding box generation and classification.
Even with five detected objects, the latency remains below 350 ms, ensuring real-time performance.
Hence,the system is suitable for dynamic environments with multiple objects.

Fig. 4. TTS Response Time vs Text Length



Text-to-speech (TTS) response time scales proportionally with input text length.
Short sentences are processed in under 200 ms, while longer descriptions take slightly longer.
The delay remains imperceptible to users, maintaining a smooth auditory experience.
This confirms that the TTS module is optimized for real-time assistive interaction.

Fig. 5. Overall System Accuracy vs Edge Model Size



System accuracy improves with larger model architectures due to better feature extraction.

The Tiny model balances low latency with moderate accuracy, while the medium model achieves peak accuracy (~95%). However, higher model complexity slightly increases power and memory consumption. Thus, the small model provides the best trade-off between speed, accuracy, and efficiency.

5. CONCLUSION AND FUTURE WORK

The proposed Edge AI–Enabled Smart Vision Assistant provides an intelligent and affordable assistive solution for visually challenged individuals by integrating multiple AI technologies into a single compact system. Through the combination of Optical Character Recognition (OCR), Text-to-Speech (TTS), real-time Object Detection, and conversational ChatGPT interaction, the device enables users to perceive, understand, and interact with their surroundings effectively. The implementation on a Raspberry Pi platform demonstrates the feasibility of deploying advanced machine learning models on low-cost edge hardware while maintaining real-time performance and privacy. The system enhances independence, accessibility, and confidence for visually impaired users by providing immediate audio feedback for both text and environmental awareness. While the current prototype successfully achieves its primary goals of text reading, object identification, and audio interaction, there remain several opportunities for enhancement. Future work can focus on improving object detection accuracy under varying lighting and outdoor conditions using more robust

deep learning models such as MobileNet-SSD. Incorporating GPS and ultrasonic sensors could enable safe outdoor navigation and obstacle avoidance. Additionally, optimizing the integration of large language models for offline or hybrid operation would reduce dependency on cloud services and latency. Expanding multilingual support in both OCR and TTS modules can further increase usability across diverse populations. Finally, developing a lightweight wearable version—such as smart glasses or a pocket-mounted unit—would improve portability and comfort, making the system a comprehensive real-world companion for the visually challenged community.

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